By including a rheostat (Rh) in the primary circuit, the experiment can be repeated several times by changing the current flowing through it.

## 2.5.7 Measurement of internal resistance of a cell by potentiometer

To measure the internal resistance of a cell, the circuit connections are made as shown in Figure 2.29. The end C of the potentiometer wire is connected to the positive terminal of the battery Bt and the negative terminal of the battery is connected to the end D through a key $K_1$. This forms the primary circuit.

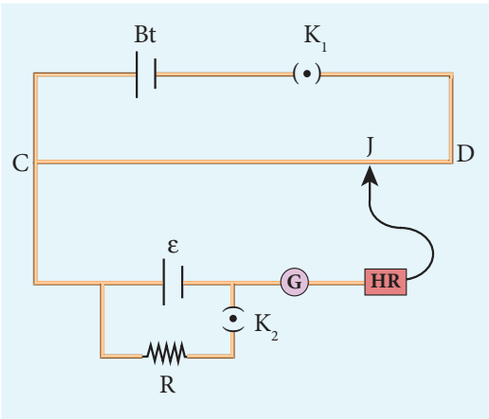



**Figure 2.29** measurement of internal resistance

The positive terminal of the cell of emf ε whose internal resistance is to be determined is also connected to the end C of the wire. The negative terminal of the cell ε is connected to a jockey through a galvanometer and a high resistance. A resistance box R and key $K_2$ are connected across the cell ε. With $K_2$ open, the balancing point J is obtained and the balancing length CJ = $l_1$ is measured. Since the cell is in open circuit, its emf is

$$\varepsilon \propto l_1 \qquad (2.62)$$

A suitable resistance (say, 10 Ω) is included in the resistance box and key $K_2$ is closed. Let r be the internal resistance of the cell. The current passing through the cell and the resistance R is given by

$$I = \frac{\varepsilon}{R+r}$$

The potential difference across R is

$$V = \frac{\varepsilon R}{R+r}$$

When this potential difference is balanced on the potentiometer wire, let $l_2$ be the balancing length.

$$\text{Then } \frac{\varepsilon R}{R+r} \propto l_2 \qquad (2.63)$$

From equations (2.62) and (2.63)

$$\frac{R+r}{R} = \frac{l_1}{l_2} \qquad (2.64)$$

$$1+\frac{r}{R} = \frac{l_1}{l_2};$$

$$r = R\left[\frac{l_1}{l_2} - 1\right]$$

$$\therefore r = R\left(\frac{l_1 - l_2}{l_2}\right) \qquad (2.65)$$

Substituting the values of the $R$, $l_1$ and $l_2$, the internal resistance of the cell is determined. The experiment can be repeated for different values of $R$. It is found that the internal resistance of the cell is not constant but increases with increase of external resistance connected across its terminals.

## 2.6 HEATING EFFECT OF ELECTRIC CURRENT

**When current flows through a resistor, some of the electrical energy delivered to the resistor is converted into heat energy and it is dissipated. This heating effect of**



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